
Interpretability of $\pi_{0.5}$ via Sparse Autoencoders

Abstract

This project investigates internal representations within the continuous flow-matching architecture of the $\pi_{0.5}$ vision-language-action model. While sparse autoencoders (SAEs) have proven effective for discrete models, we explore their applicability to continuous generation systems. By training SAEs on high-dimensional activations from the $\pi_{0.5}$ semantic-action interface, we aim to identify whether semantic and physical concepts can be decoupled within a continuous-time vector field. Our analysis seeks to provide preliminary insights into the internal mechanisms and feature interpretability of flow-based embodied models.

1 Team

The research team consists of four members: Xin Gao (高鑫), Xinyue Li (李馨月), Zichen Wen (温子辰), Honglin Lin (林泓霖). The members' research areas encompass large language, vision-language, and video generation models, directly motivating our focus on interpreting the internal representations of VLA systems.

2 Plan

2.1 Background and Motivation

Conventional vision-language-action architectures rely on autoregressive discrete decoding, which introduces latency during high-frequency control [2]. To address this, the $\pi_{0.5}$ model integrates a vision-language backbone with a continuous flow-matching action expert. This architecture employs a knowledge insulation mechanism—truncating continuous gradients to prevent the degradation of generalized semantics during co-training [1]. Despite its empirical success, the internal division of labor within this hybrid system remains opaque. While sparse autoencoders (SAEs) offer a proven framework for extracting interpretable features from discrete networks [3], applying them to the time-dependent non-linearity of continuous flow matching introduces novel representational complexities. This research addresses this gap by analyzing activations at the $\pi_{0.5}$ insulated interface to evaluate feature decoupling within continuous-time vector fields and explore their functional roles.

2.2 Specific Aims

Aim 1 (Activation Extraction). Deploy forward hooks in $\pi_{0.5}$ during simulated tasks to isolate high-dimensional activations at the interface between the vision-language backbone and the flow-matching solver.

Aim 2 (Sparse Dictionary Learning). Train Top- K sparse autoencoders on the extracted continuous activations using reconstruction and auxiliary objectives to produce sparse feature dictionaries.

Aim 3 (Qualitative Feature Analysis). Analyze the learned dictionaries by evaluating top-activating frames and trajectories to identify interpretable semantic and physical features.

Aim 4 (Exploratory Functional Assessment). Conduct preliminary feature interventions by applying localized steering perturbations to specific dictionary directions to assess nonlinear stability.

2.3 Related Work

The evolution of embodied foundation models demonstrates a transition from discrete tokenization to continuous generation architectures. The OpenVLA framework exemplifies the autoregressive

discrete paradigm, training instruction-conditioned policies while maintaining actions within a tokenized space [2]. To address the efficiency and precision limits of discrete approaches, the $\pi_{0.5}$ model adopts a continuous flow-matching architecture integrated with a semantic backbone [1]. Notably, $\pi_{0.5}$ implements a knowledge insulation mechanism to separate discrete adaptation from the continuous expert, providing the specific structural context for this investigation. Concurrently, advancements in mechanistic interpretability show that sparse autoencoders (SAEs) can successfully isolate transferable features and facilitate causal steering within policy networks [3]. This study synthesizes these trajectories by adapting established SAE methodologies to evaluate the continuous generation interface of the $\pi_{0.5}$ architecture.

2.4 Timeline

Weeks 1 to 2. Deploy the simulation environment and establish a data extraction pipeline to deterministically log dense network representations.

Weeks 3 to 4. Train sparse autoencoders on the extracted activations to stabilize feature dictionaries. Principal component analysis provides a fallback strategy against optimization instability.

Weeks 5 to 6. Qualitatively evaluate learned dictionaries to identify interpretable physical concepts and execute preliminary feature interventions to assess nonlinear stability.

Week 7. Synthesize experimental results and formalize qualitative findings regarding functional variations and continuous feature decoupling.

Week 8. Compile the final academic report including visual panels and a supplementary appendix to ensure methodology reproducibility.

2.5 Roles and Division of Labor

All four members make equal contributions, collaboratively participating in core coding, evaluation, and reporting. The precise division of labor will dynamically adapt to experimental developments and practical requirements.

- **林泓霖 (System Integration).** Implements the simulation environments and constructs the codebase for the $\pi_{0.5}$ activation extraction infrastructure.
- **温子辰 (Algorithm Implementation).** Leads the implementation of the sparse autoencoder architecture and guides the development of loss functions to ensure training stability.
- **高鑫 (Interpretability Analysis).** Performs the qualitative evaluation of the extracted feature dictionaries to continuously identify and verify decoupled physical concepts.
- **李馨月 (Research Formulation and Functional Assessment).** Formulates primary research objectives and programs the feature steering experiments to evaluate nonlinear robustness.

References

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